Marine Microbial Biodiversity

Lecture 04

Microbial Metabolism Pt.2

Matteo Selci, PhD

matteo.selci@unina.it

www.donatogiovannelli.com

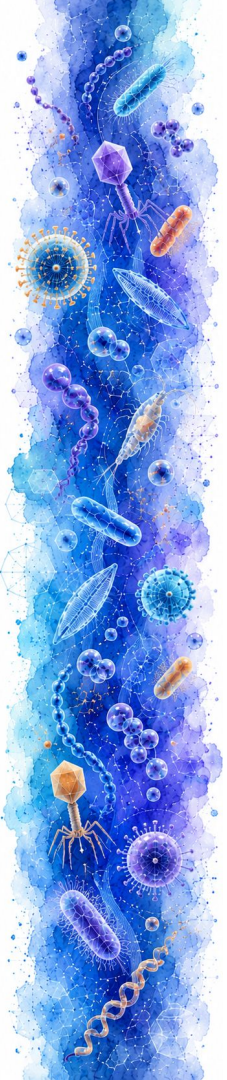
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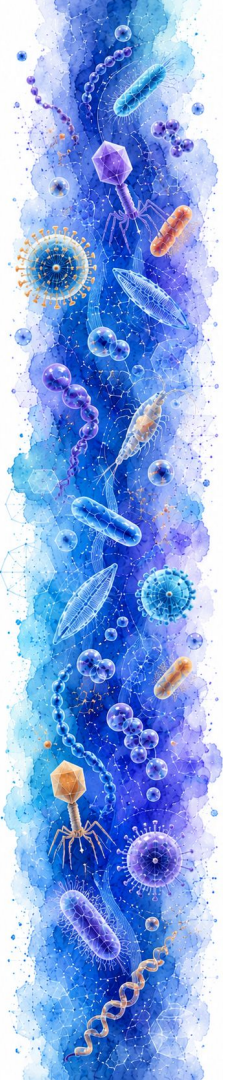
Chemotrophs

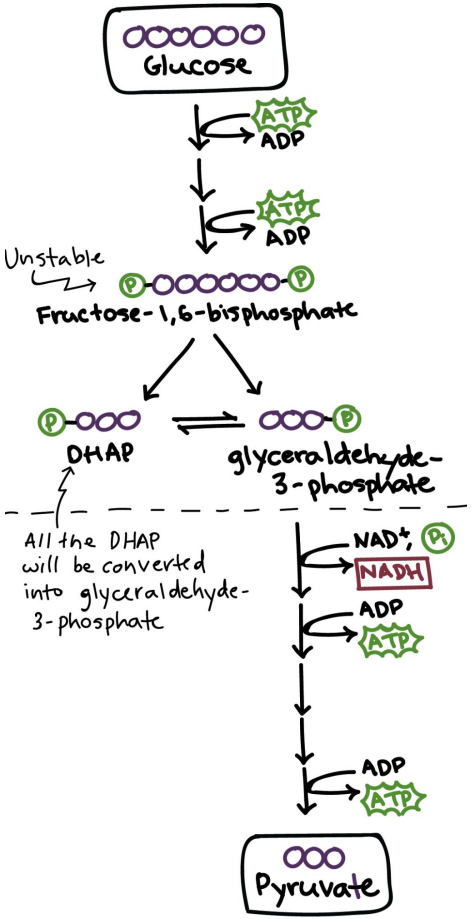
Chemotrophs are organisms that obtain energy through a chemical process called chemosynthesis, rather than by photosynthesis. In chemotrophs, **energy is obtained by chemicals**, organic or inorganic, throughout redox reactions that transfer electrons from an initial electron donor to a final electron acceptor.

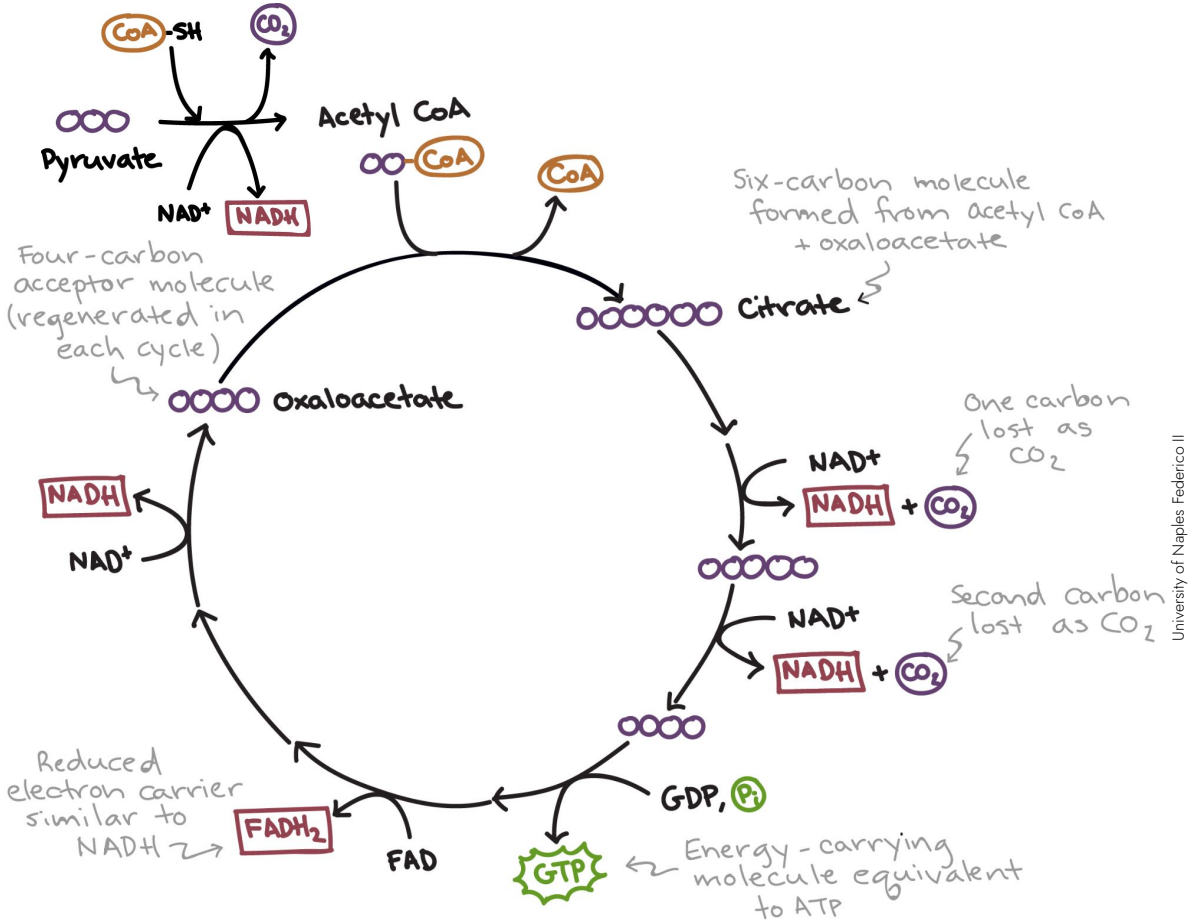
Chemoorganotrophy is a term used to denote the **oxidation of organic chemicals** to yield energy. Chemoorganotrophs are by definition also heterotrophs. They perform catabolism of organic compounds through glycolysis, which could proceed in presence of O_2 with the citric acid cycle and the oxidative phosphorylation, a pathway known as **aerobic respiration**. They use organic compounds both as carbon source and electron donor, and oxygen as terminal electron acceptor.

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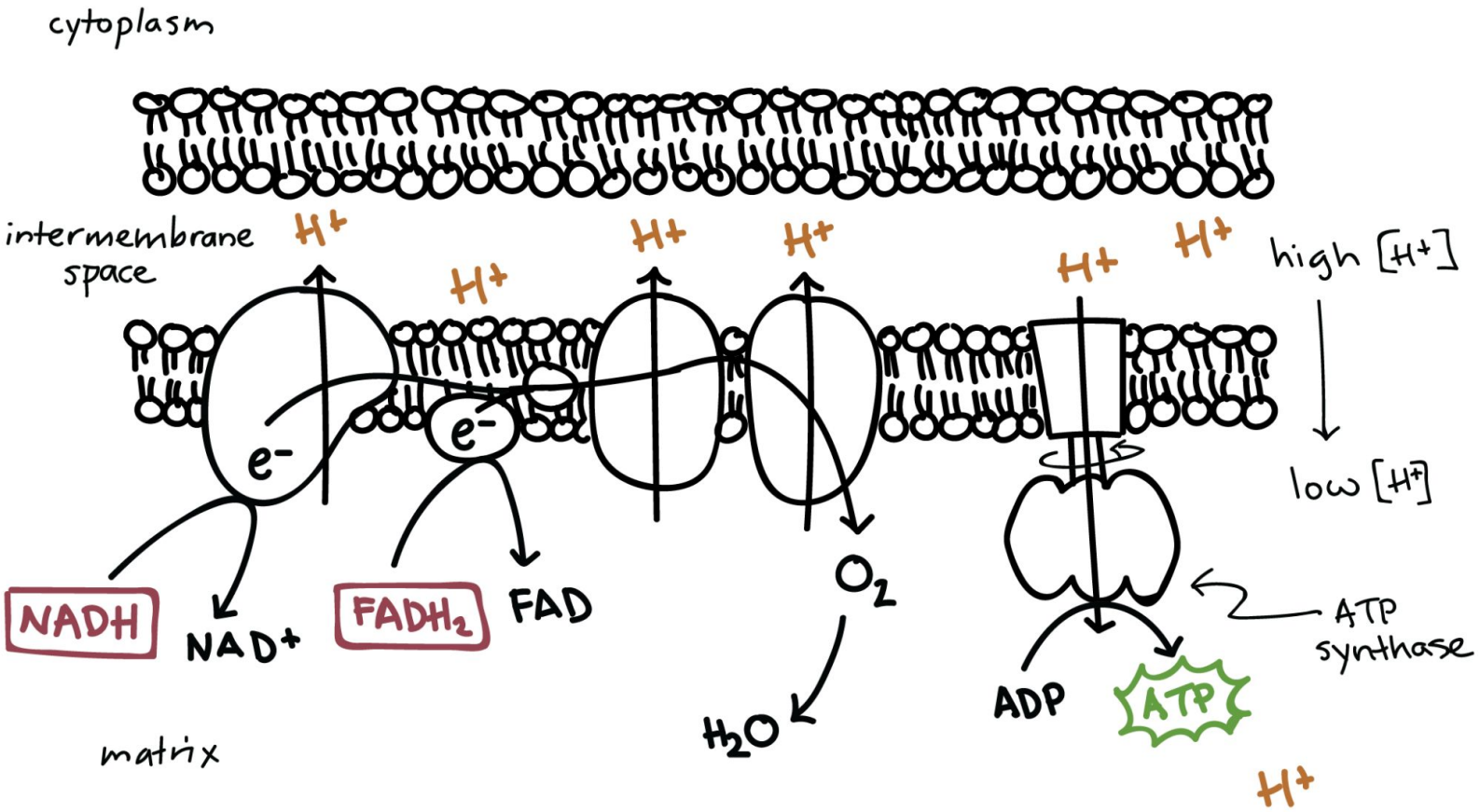






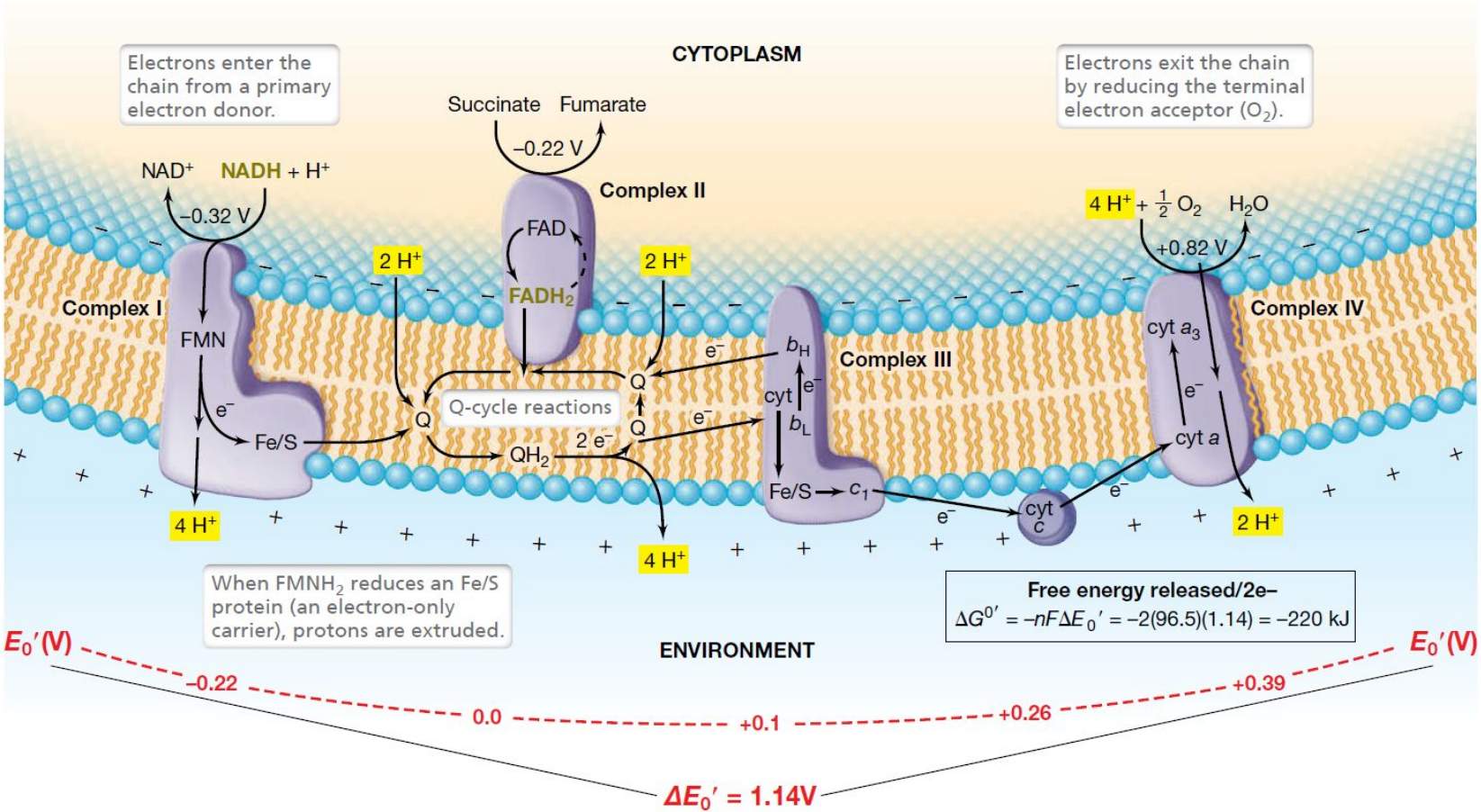
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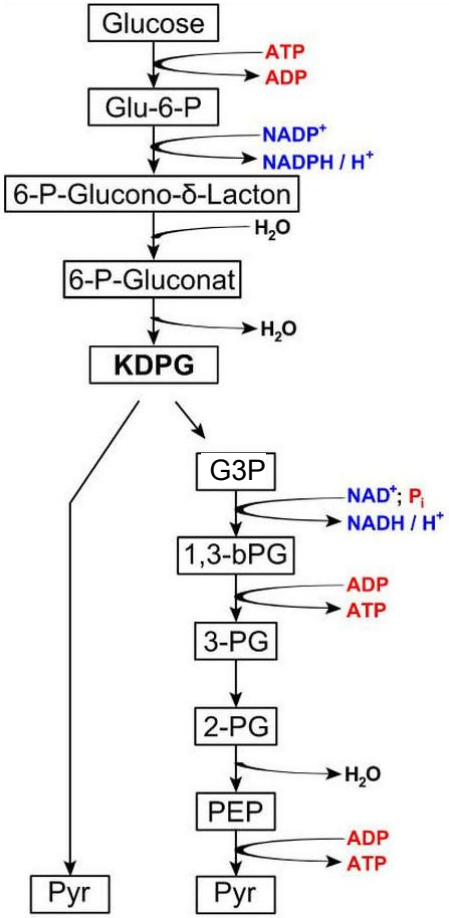
Aerobic Respiration



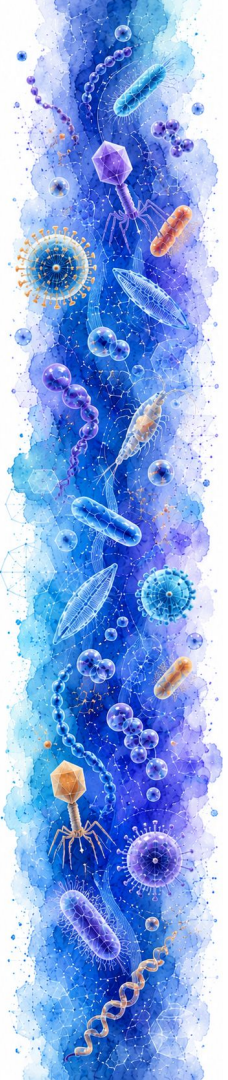
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Entner-Doudoroff - Alternative Glucose Oxidation Pathway

The Entner-Doudoroff pathway is an **alternative glycolytic pathway** that enables the breakdown of glucose into pyruvate. It is found in a variety of microorganisms, including **some bacteria** and **archaea that lack key enzymes of the classical glycolytic route**.
This pathway begins similarly to glycolysis with the phosphorylation of glucose into glucose-6-phosphate (G6P) using ATP. Then, G6P is oxidized to 6-phosphogluconolactone, reducing $NADP^+$ to NADPH, which is then hydrolyzed to 6-phosphogluconate. 6-phosphogluconate undergoes dehydration by the enzyme 6-phosphogluconate dehydratase, forming 2-keto-3-deoxy-6-phosphogluconate (KDPG). KDPG, is cleaved by KDPG aldolase into two important products: pyruvate and glyceraldehyde-3-phosphate (G3P). Starting from this step G3P is processed via the same steps as in glycolysis (the EMP pathway), ultimately producing another molecule of pyruvate, as well as ATP and NADH. **The Entner-Doudoroff pathway produces 1 ATP, 1 NADH, and 1 NADPH per molecule of glucose.**



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Chemotrophs

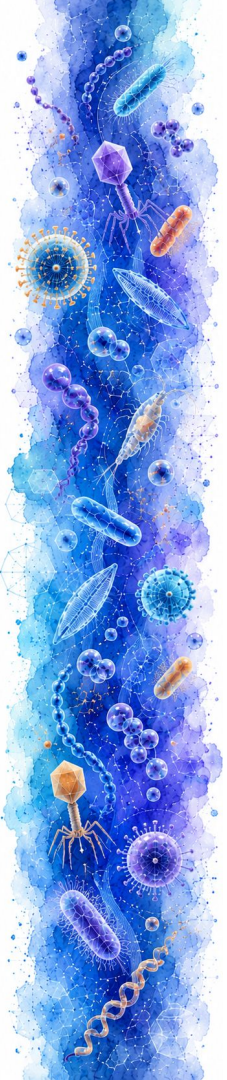
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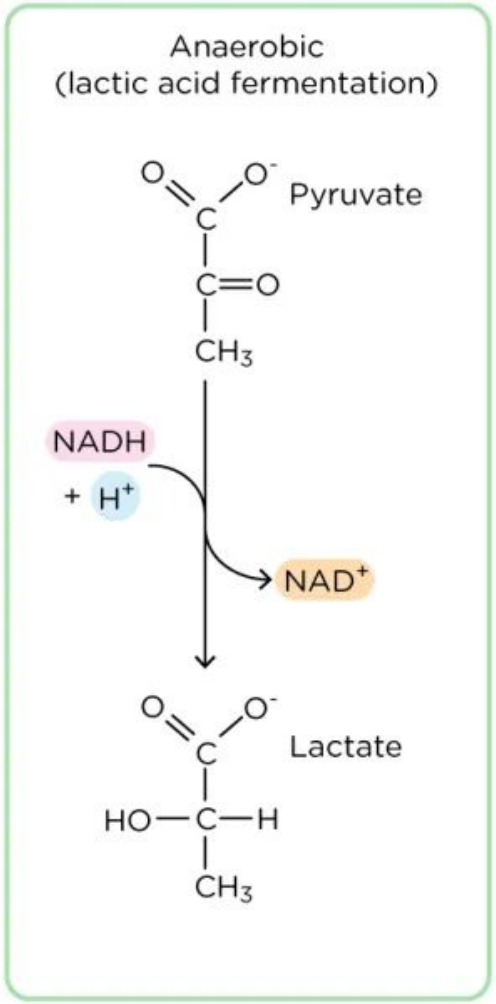
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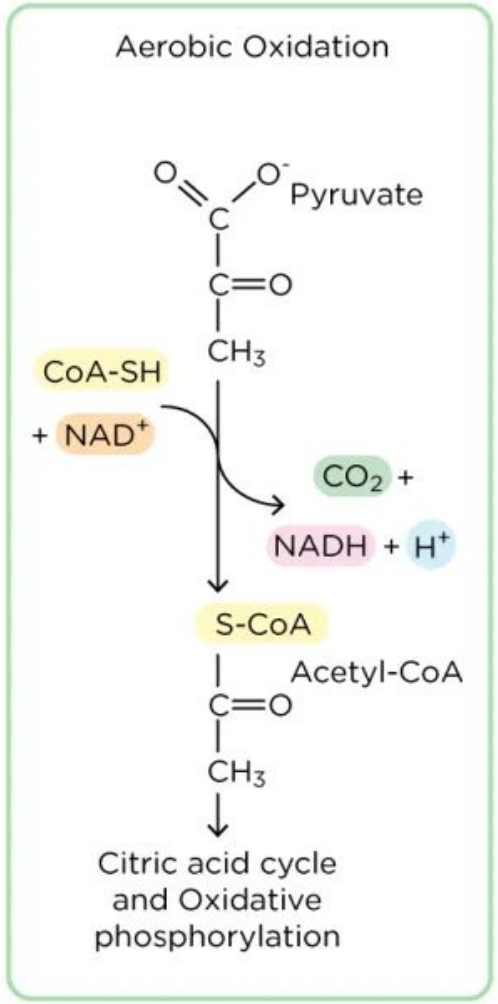
Under **anoxic** conditions, **inorganic electron acceptors** other than O_2 support respiration in a wide variety of microbes in the process called **anaerobic respiration**: ferric iron (Fe^{3+}), nitrate (NO_3^-), and nitrite (NO_2^-), sulfate (SO_4^{2-}), elemental sulfur (S^0), and carbon dioxide (CO_2) among others.

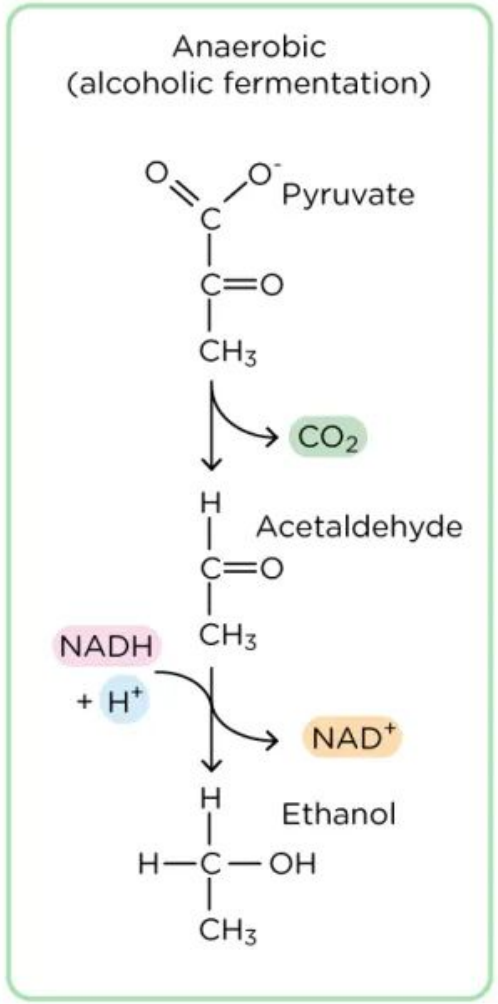
As a third option, **fermentation** is a form of **anaerobic catabolism** in which organic compounds both donate and accept electrons, and redox balance is achieved without the need for external electron acceptors, generating ATP as a result of substrate-level phosphorylation, and not by harnessing an ion motive force.

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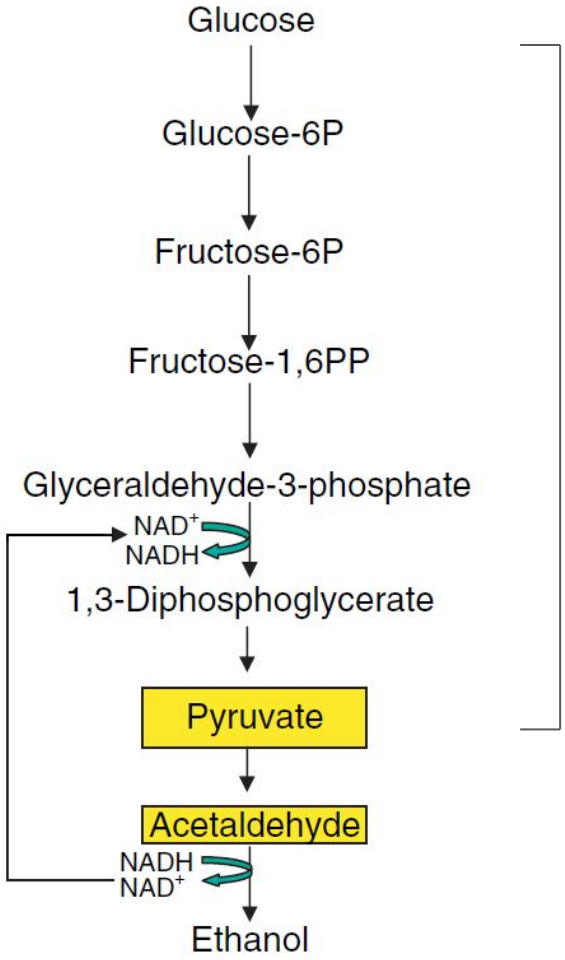




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Alcoholic Fermentation

Alcoholic fermentation is carried out by **yeasts** and some other **fungi** and **bacteria**. The first step of the alcoholic fermentation pathway involves pyruvate, which is decarboxylated to acetaldehyde in a reaction that is catalyzed by the enzyme pyruvate decarboxylase.
The **redox balance** of alcoholic fermentation is achieved by the regeneration of NAD+ during the reduction of acetaldehyde to ethanol, which is catalyzed by alcohol dehydrogenase.
The ATP yield of alcoholic fermentation is generally 2 mol of ATP per mole of glucose.



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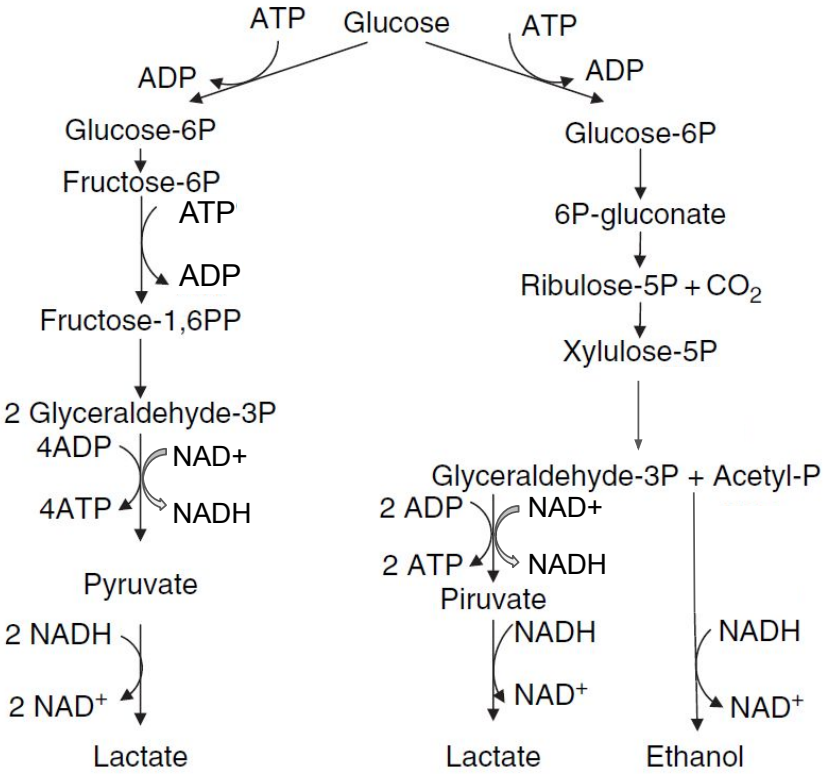
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Lactic Acid Fermentation

Lactic acid fermentation is carried out by lactic acid bacteria and bifidobacteria, and also by some species of Bacillus, some protozoa and water molds.

Homofermentative: convert sugar to lactic acid via glycolysis. The enzyme lactate dehydrogenase leads to pyruvate reduction and NADH reoxidation, with the **production of lactate** (single product). ATP yield is **2** mol per **1** mole of glucose oxidized.

Heterofermentative: do not perform glycolysis due to the lack of aldolase, Glucose 6-phosphate is oxidized to 6-phosphogluconate and fermentation is carried out through the phosphoketolase pathway, leading to the **production of lactate**, **ethanol**, and **CO_2** (more products). ATP yield is **1** mol per **1** mole of glucose.

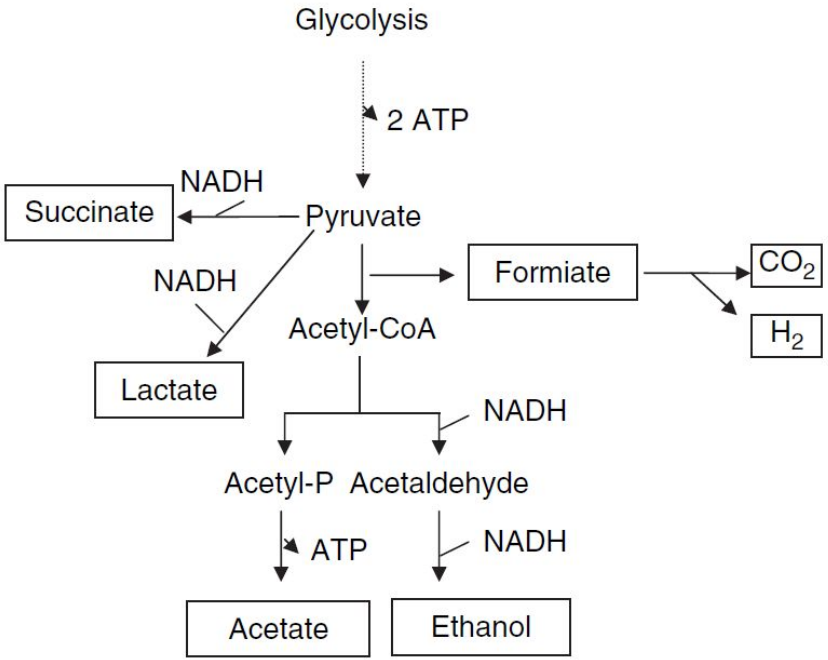


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Mixed-Acid Fermentation

Mixed-acid fermentation is characteristic of the **Enterobacteriaceae** and to some species of **Aeromonas**; it is also carried out by some anaerobic fungi. It involves the **fermentation** of monosaccharides, disaccharides, polyalcohol, and, less frequently, polysaccharides, **via the glycolytic pathway**, **producing lactic**, **formic**, **succinic** and **acetic acids**, and **ethanol**. The final amounts of each end product vary depending on the microorganism and the growth conditions; however, the **ratio of acid to neutral products is 4:1**.
ATP yield of mixed-acid fermentation is about **3** mol of ATP per **1** mole of glucose.



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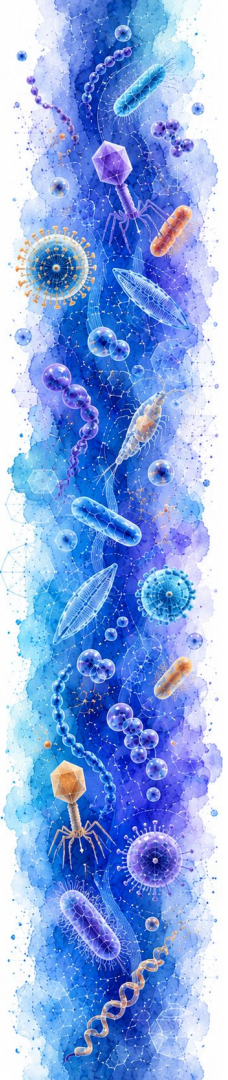


TABLE 14.8 Common fermentations and their energetics and example organisms

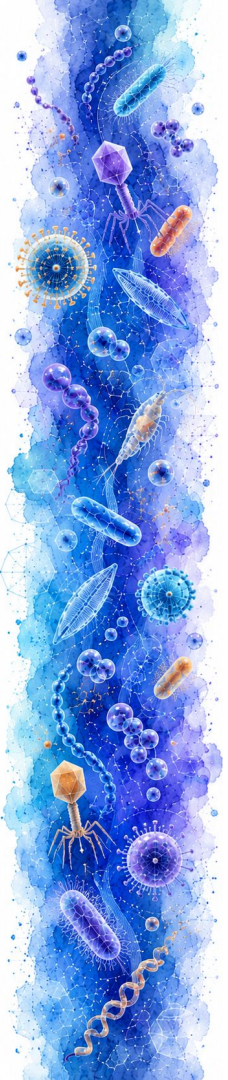
Type	*Reaction*	*Energy yield ($\Delta G^{0\prime}$, kj/mol)*	*Organisms*
Alcoholic	Hexose $\rightarrow$ 2 ethanol + 2 CO_2	−239	Yeast, *Zymomonas*
Homolactic	Hexose $\rightarrow$ 2 lactate$^-$ + 2 H^+	−196	*Streptococcus,* some *Lactobacillus*
Heterolactic	Hexose $\rightarrow$ lactate$^-$ + ethanol + CO_2 + H^+	−216	*Leuconostoc,* some *Lactobacillus*
Propionic acid	3 Lactate$^-$ $\rightarrow$ 2 propionate$^-$ + acetate$^-$ + CO_2 + H_2O	−170	*Propionibacterium, Clostridium propionicum*
Mixed acid	Hexose $\rightarrow$ ethanol + 2,3-butanediol + succinate^{2-} + lactate$^-$ + acetate$^-$ + formate$^-$ + H_2 + CO_2	Depends on product ratio	Enteric bacteria including *Escherichia, Salmonella, Shigella, Klebsiella, Enterobacter*
Butyric acid	Hexose $\rightarrow$ butyrate$^-$ + 2 H_2 + 2 CO_2 + H^+	−264	*Clostridium butyricum*
Butanol	2 Hexose $\rightarrow$ butanol + acetone + 5 CO_2 + 4 H_2	−468	*Clostridium acetobutylicum*
Caproate/ Butyrate	6 Ethanol + 3 acetate$^-$ $\rightarrow$ 3 butyrate$^-$ + caproate$^-$ + 2 H_2 + 4 H_2O + H^+	−183	*Clostridium kluyveri*
Acetogenic	Fructose $\rightarrow$ 3 acetate$^-$ + 3 H^+	−276	*Clostridium aceticum*

REMINDER

The tendency of a substance to donate or accept electrons is defined by its reduction potential (E_0'). In redox reactions, the larger the difference in reduction potential between electron donor and electron acceptor, the more free energy is released ($\Delta G^{0\prime}$).

Glycolysis $\Delta G^{0\prime}$= -686 kJ/mol

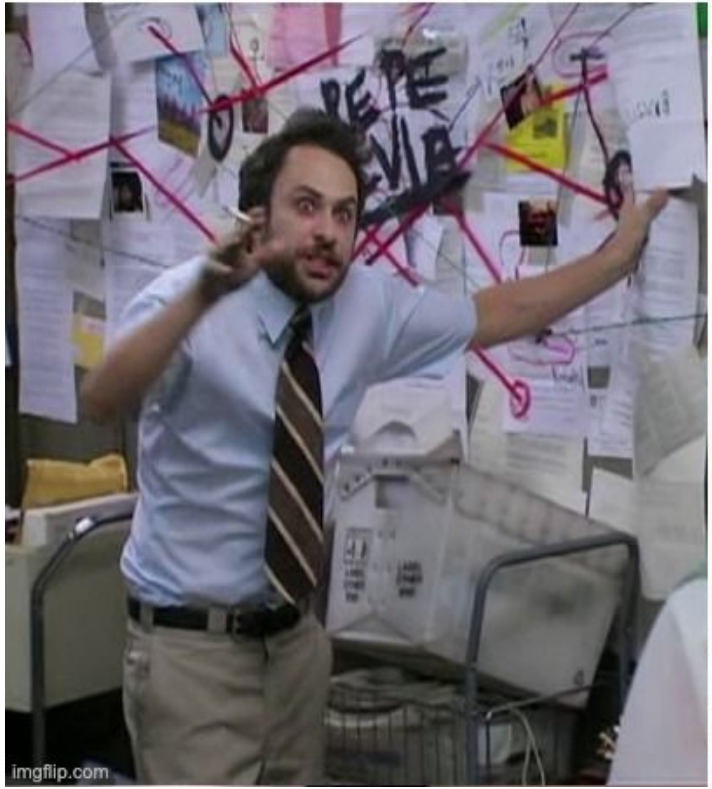
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Common Fermentation Pathways			
Pathway	**End Products**	**Example Microbes**	**Commercial Products**
Acetone-butanol-ethanol	Acetone, butanol, ethanol, CO_2	*Clostridium acetobutylicum*	Commercial solvents, gasoline alternative
Alcohol	Ethanol, CO_2	*Candida, Saccharomyces*	Beer, bread
Butanediol	Formic and lactic acid; ethanol; acetoin; 2,3 butanediol; CO_2; hydrogen gas	*Klebsiella, Enterobacter*	Chardonnay wine
Butyric acid	Butyric acid, CO_2, hydrogen gas	*Clostridium butyricum*	Butter
Lactic acid	Lactic acid	*Streptococcus, Lactobacillus*	Sauerkraut, yogurt, cheese
Mixed acid	Acetic, formic, lactic, and succinic acids; ethanol, CO_2, hydrogen gas	*Escherichia, Shigella*	Vinegar, cosmetics, pharmaceuticals
Propionic acid	Acetic acid, propionic acid, CO_2	*Propionibacterium, Bifidobacterium*	Swiss cheese

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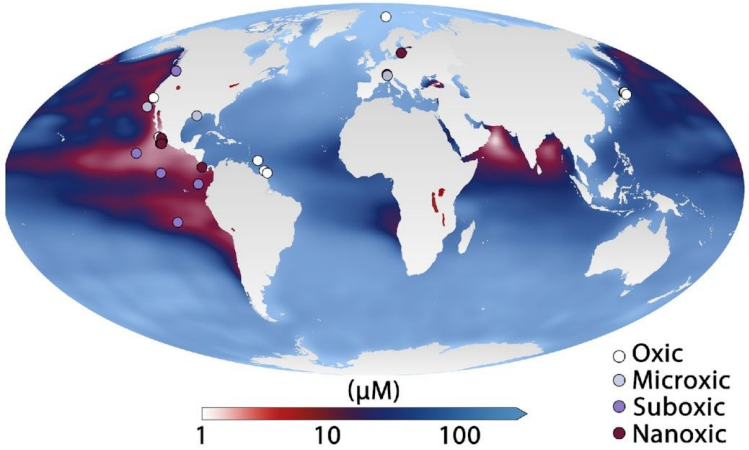
ME EXPLAINING
BACTERIAL METABOLISM
AND BIOSYNTHETIC PATHWAYS



LITERALLY EVERYONE ELSE

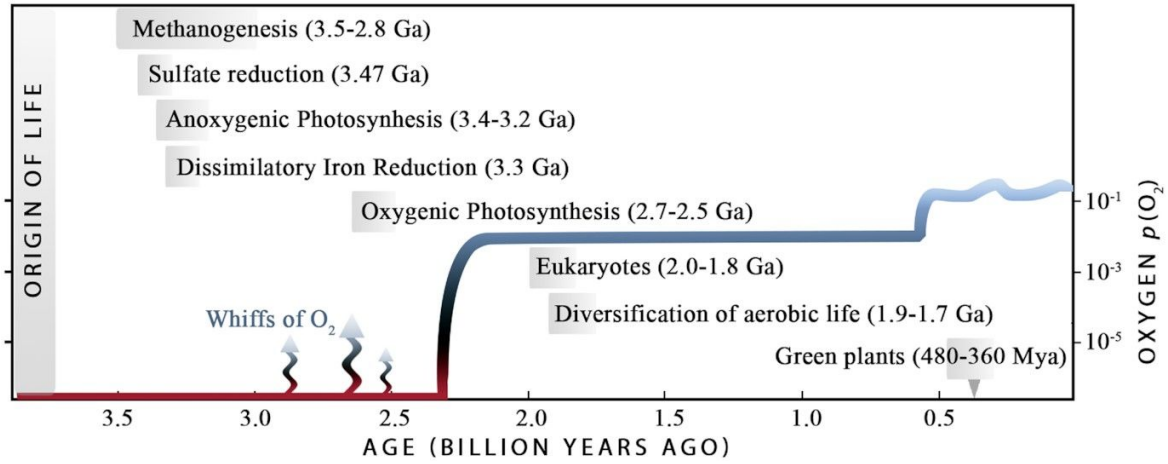


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World map of minimum O_2 concentrations in aquatic environments

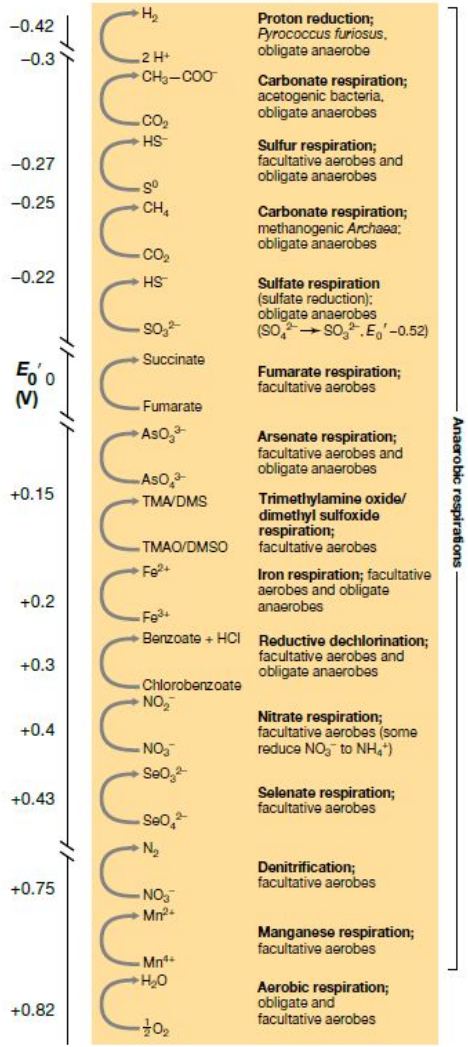
Biogeological timeline showing the evolution of microbial metabolisms in parallel with the rise in atmospheric oxygen concentrations.



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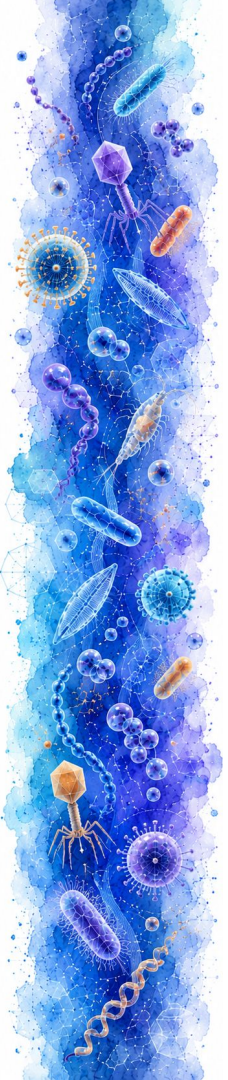
Anaerobic Respiration

Organic compounds	Inorganic compounds
Fumarate	Nitrate (NO_3^-)
Dimethylsulfoxide (DMSO)	Nitrite (NO_2^-)
Trimethylamine *N*-oxide (TMAO)	Nitrous oxide (N_2O)
	Chlorate (ClO_3^-)
	Perchlorate (ClO_4^-)
	Manganic ion (Mn^{4+})
	Ferric ion (Fe^{3+})
	Gold (Au^{3+})
	Selenate (SeO_4^{2-})
	Arsenate (AsO_4^{3-})
	Sulfate (SO_4^{2-})
	Sulfur (S^0)



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Anaerobic Respiration

In microbial metabolism, dissimilative and assimilative pathways represent two different modes of utilizing substances, typically inorganic compounds, and they serve distinct biological purposes.

ASSIMILATORY

- inorganic compounds like nitrate (NO_3^-), sulfate (SO_4^{2-}), or carbon dioxide (CO_2) are reduced for biosynthetic purposes
- energy is consumed
- only a small amount of compounds is reduced
- nitrogen, sulfur, or carbon are assimilated and converted into organic compounds
- conserved pathways, common in both autotrophs and heterotrophs

DISSIMILATORY

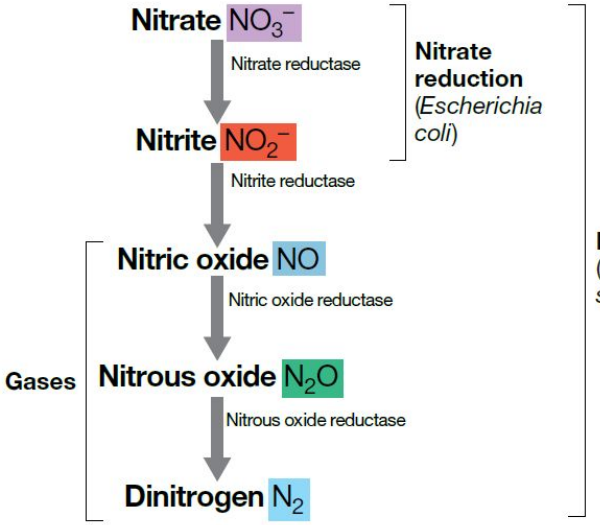
- inorganic compounds like nitrate (NO_3^-), sulfate (SO_4^{2-}), or carbon dioxide (CO_2) are reduced for energetic purposes
- energy is gained
- a large amount of the electron acceptor is reduced
- reduced nitrogen, sulfur, or carbon forms are excreted as waste products
- pathways only found in prokaryotic group capable of anaerobic respiration

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Anaerobic Respiration: Nitrate Reduction and Denitrification

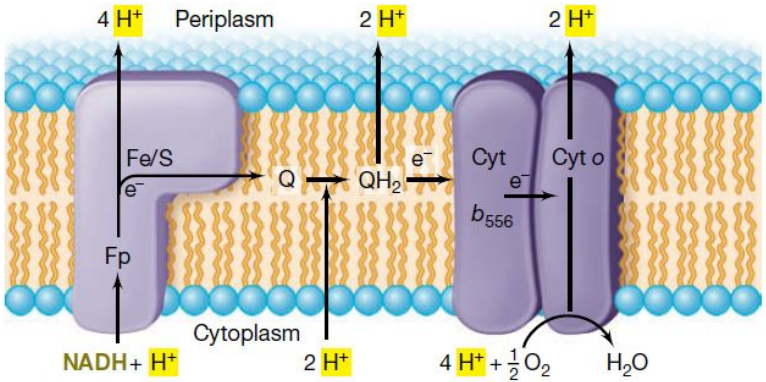
Inorganic nitrogen compounds are some of the most common electron acceptors in anaerobic respiration.

One of the most common alternative electron acceptors is **nitrate** (NO_3^-), in a pathway called dissimilatory reduction of nitrate, which can be reduced with two electrons to **nitrite** (NO_2^-), or reduced further to **nitric oxide** (NO), **nitrous oxide** (N_2O), or **dinitrogen** (N_2).
Because **NO, N_2O, and N_2 are all gases**, they can be lost from the marine environment, and their biological production is called denitrification.
Furthermore, the intermediate gases have environmental consequences in climate warming and production of acid rain, so these processes are of critical importance in marine nitrogen cycling, especially in anoxic environments and oxygen minimum zones.

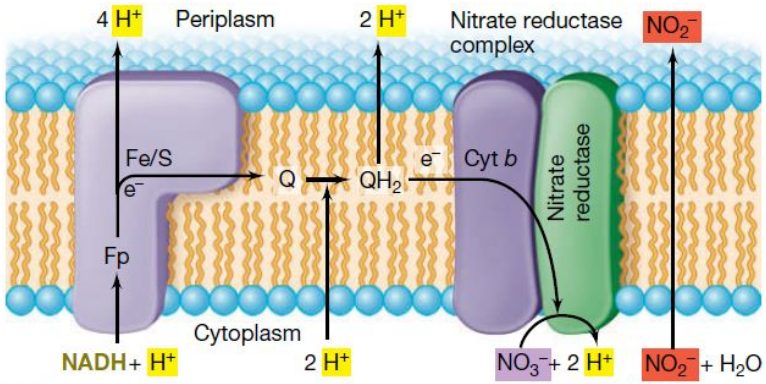


Pathway of dissimilatory reduction of nitrate

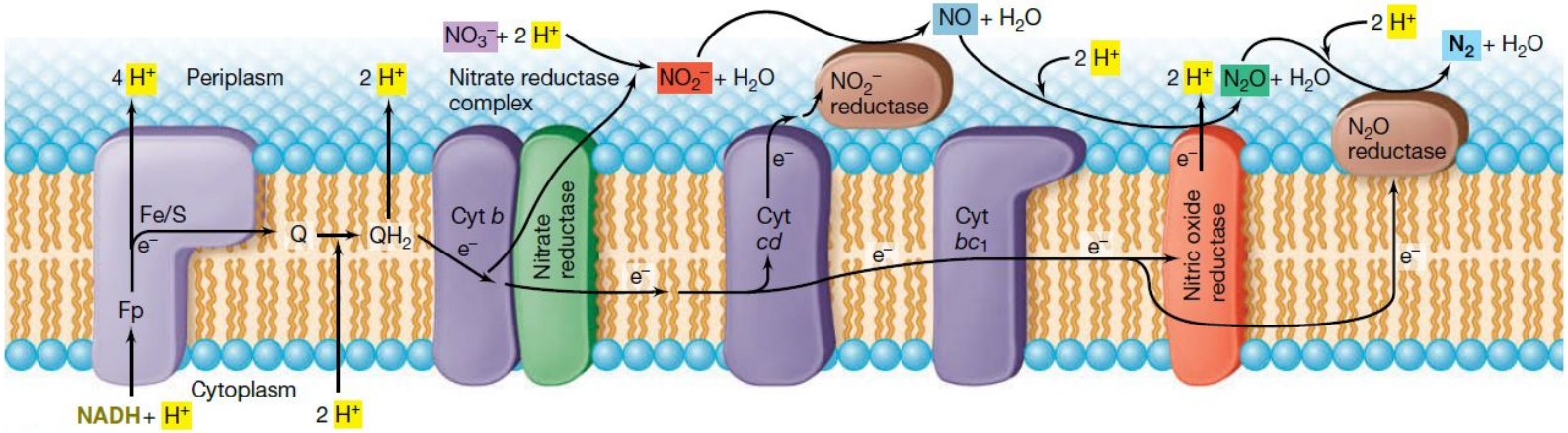
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(a) **Aerobic respiration**



(b) **Nitrate reduction**



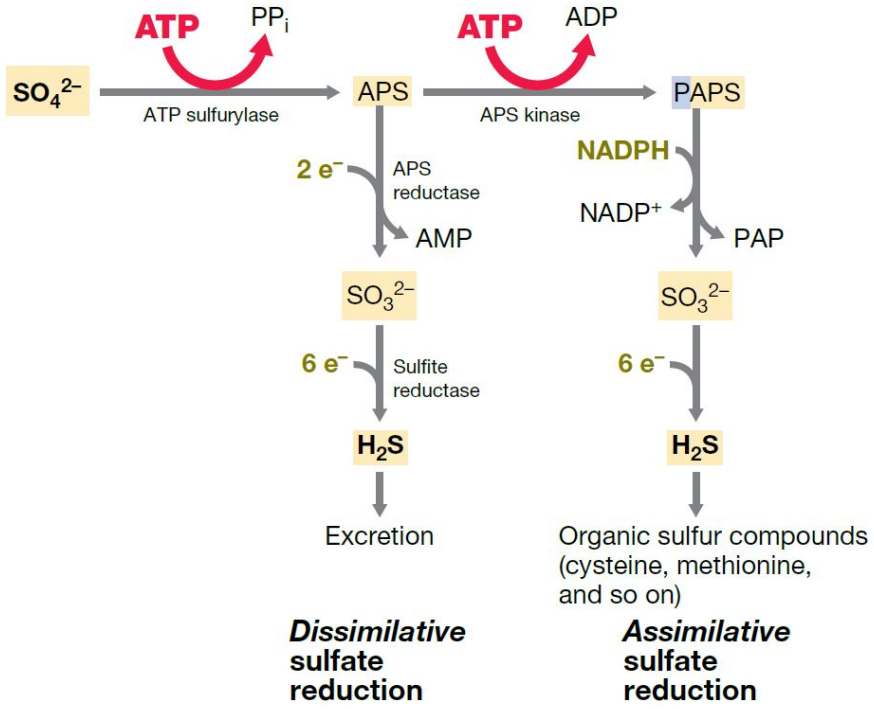
(c) **Denitrification**

Paracoccus denitrificans and *Pseudomonas stutzeri*

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Anaerobic Respiration: Sulfate Reduction

Several inorganic sulfur compounds are important electron acceptors in anaerobic respiration.
Sulfate (SO_4^{2-}), the most oxidized form of sulfur, is reduced by the **sulfate-reducing bacteria**, with the exception of Archaeoglobus, a genus of Archaea.
Most microbes can **incorporate** sulfate for biosynthetic purposes to make cysteine, methionine, and many other organosulfur compounds; this is assimilative sulfate metabolism (fixation).
By contrast, the ability to use sulfate as an **electron acceptor** for **energy conservation**, or dissimilative sulfate reduction (respiration) is restricted to the sulfate-reducing bacteria.

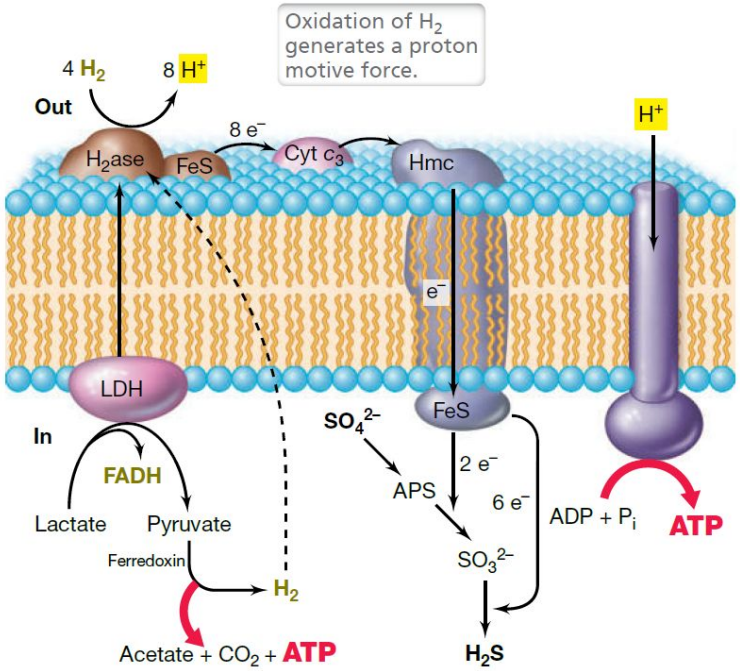


Pathways of assimilatory and dissimilatory reduction of sulfate

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Anaerobic Respiration: Sulfate Reduction

SO_4^{2-} is an energetically much less favorable electron acceptor than is O_2 or NO_3^-.
However, sufficient free energy to make ATP available from sulfate reduction is possible when an additional electron donor is oxidized, yielding NADH or FADH.

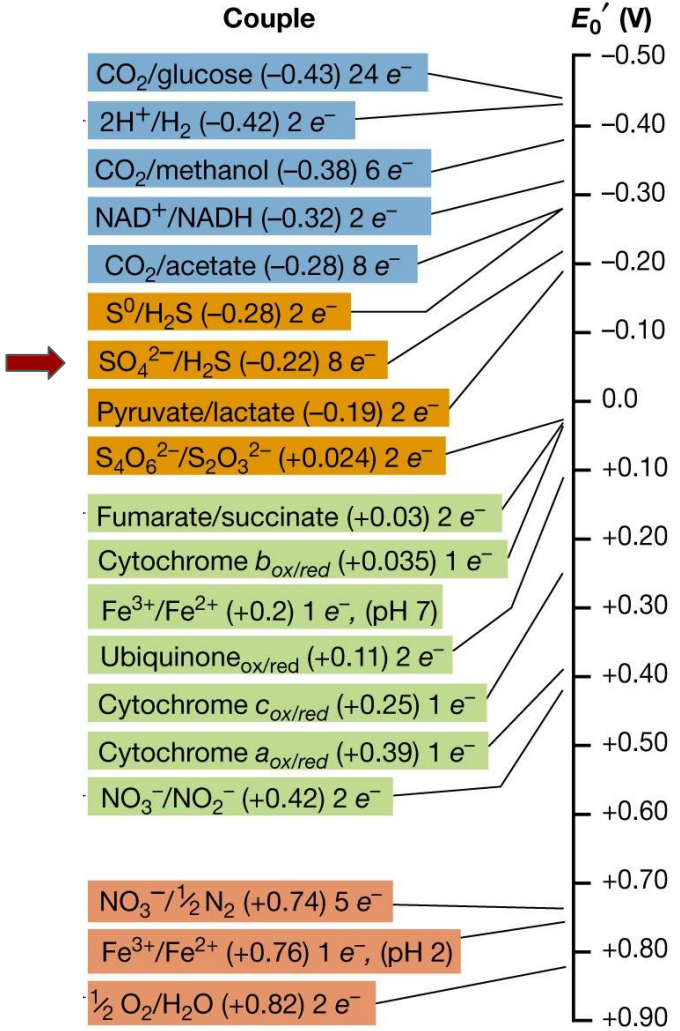


Desulfovibrio desulfuricans

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Anaerobic Respiration: Sulfate Reduction

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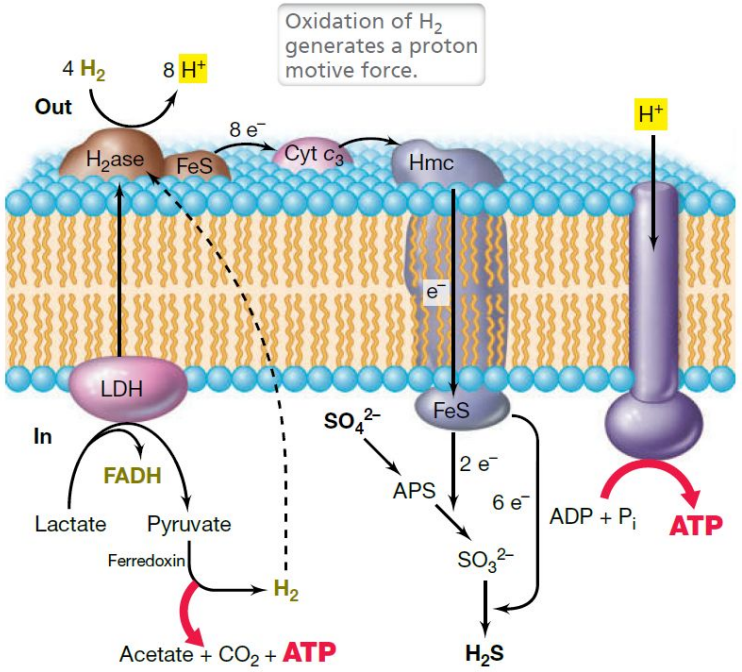


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Anaerobic Respiration: Sulfate Reduction

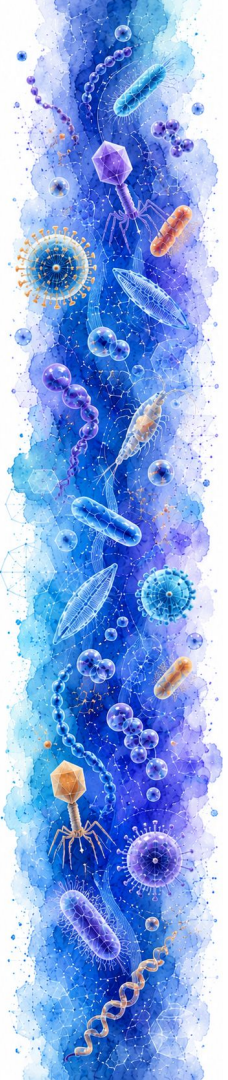
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Hydrogen (H_2) is used by virtually all species.
This H_2 crosses the cytoplasmic membrane and is oxidized by the periplasmic hydrogenase to electrons, which are fed back into the system, and protons, which establish the proton motive force.
A net of 1 ATP is produced for each SO_4^{2-} reduced to HS^-.
When lactate or pyruvate is the electron donor, ATP is produced not only from the proton motive force but also by substrate-level phosphorylation during the oxidation of pyruvate to acetate plus CO_2.



Desulfovibrio desulfuricans

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Chemotrophs are organisms that obtain energy through a chemical process called chemosynthesis, rather than by photosynthesis. In chemotrophs, **energy is obtained by chemicals**, organic or inorganic, throughout redox reactions that transfer electrons from an initial electron donor to a final electron acceptor.

Chemolithotrophy is a term used to denote the **oxidation of inorganic chemicals** to yield energy. The energetics of chemolithotrophy depend on the free-energy yield of coupled redox reactions between an electron donor and electron acceptor. They can be both autotrophs and heterotrophs, but most are **autotrophs**. They synthesize new organic matter from inorganic carbon (CO_2), therefore known as *"primary producers"*, mostly throughout the **Calvin cycle**, the major biochemical pathway by which autotrophic organisms incorporate CO_2 into cell material (CO_2 fixation).

Chemolithotrophs can tap into many natural sources of **inorganic electron donors**, including geological, biological, and anthropogenic. Electron donors with sufficient **reducing power** to drive ATP synthesis include elemental sulfur (S^0), hydrogen sulfide (H_2S), hydrogen (H_2), ferrous iron (Fe^{2+}), ammonia (NH_3), nitrite (NO_2^-), methane (CH_4), among others.

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TABLE 14.2 **Energy yields from the oxidation of various inorganic electron donors**[a]

Electron donor	*Chemolithotrophic reaction*	*Group of chemolithotrophs*	*E_0' of couple (V)*	*$\Delta G^{0\prime}$ (kJ/reaction)*	*Number of electrons/ reaction*	*$\Delta G^{0\prime}$ (kJ/2 e^-)*
Phosphite[b]	$4\ HPO_3^{2-} + SO_4^{2-} + H^+ \rightarrow 4\ HPO_4^{2-} + HS^-$	Phosphite bacteria	−0.69	−364	8	−91
Hydrogen[b]	$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$	Hydrogen bacteria	−0.42	−237.2	2	−237.2
Sulfide[b]	$HS^- + H^+ + \frac{1}{2}O_2 \rightarrow S^0 + H_2O$	Sulfur bacteria	−0.27	−209.4	2	−209.4
Sulfur[b]	$S^0 + 1\frac{1}{2}O_2 + H_2O \rightarrow SO_4^{2-} + 2\ H^+$	Sulfur bacteria	−0.20	−587.1	6	−195.7
Ammonium[c]	$NH_4^+ + 1\frac{1}{2}O_2 \rightarrow NO_2^- + 2\ H^+ + H_2O$	Nitrifying bacteria	+0.34	−274.7	6	−91.6
Nitrite[b]	$NO_2^- + \frac{1}{2}O_2 \rightarrow NO_3^-$	Nitrifying bacteria	+0.43	−74.1	2	−74.1
Ferrous iron[b]	$Fe^{2+} + H^+ + \frac{1}{4}O_2 \rightarrow Fe^{3+} + \frac{1}{2}H_2O$	Iron bacteria	+0.77	−32.9	1	−65.8

REMINDER

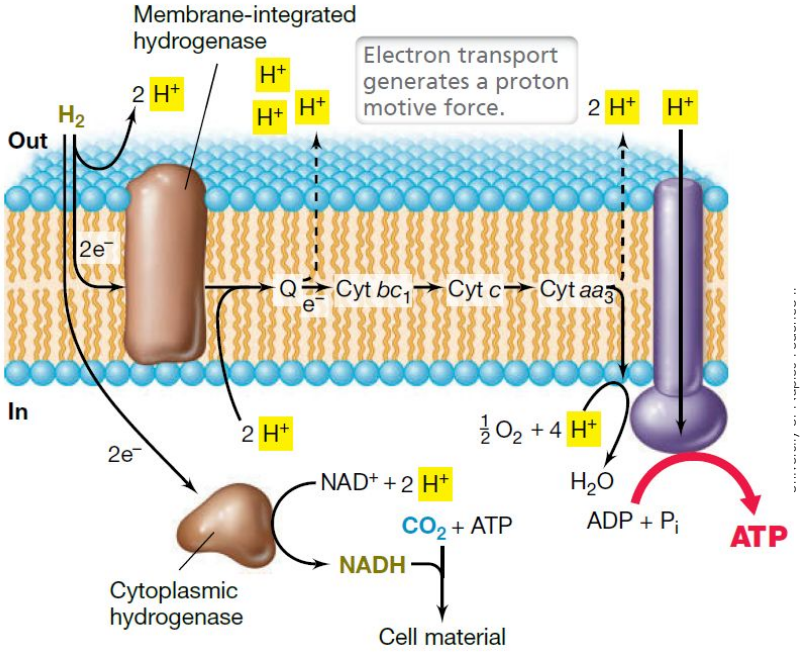
The tendency of a substance to donate or accept electrons is defined by its reduction potential (E_0'). In redox reactions, the larger the difference in reduction potential between electron donor and electron acceptor, the more free energy is released ($\Delta G^{0\prime}$).

Sulfur Oxidation $\Delta G^{0\prime}$= from -209.4 kJ/mol to -195.7 kJ/mol

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Hydrogen Oxidation

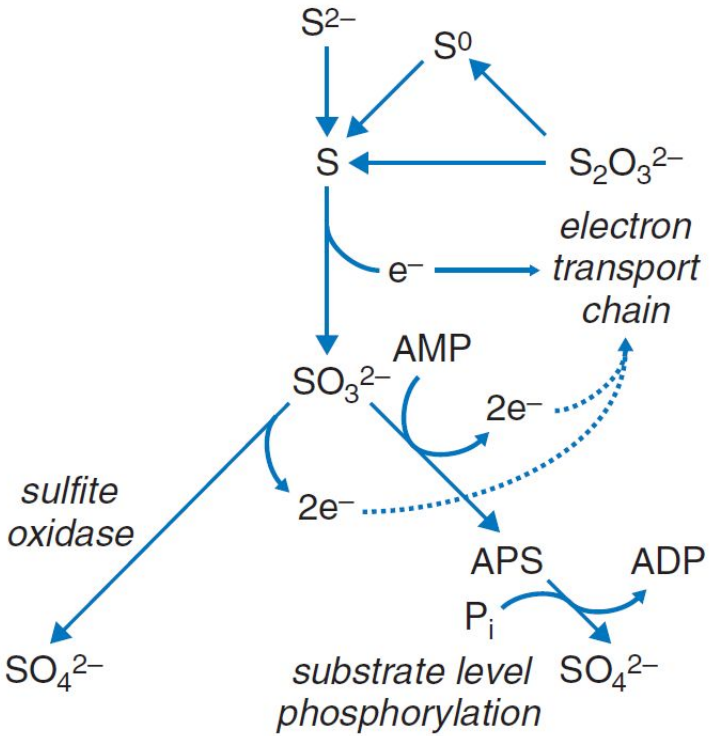
Hydrogen (H_2) is a good electron donor, which can give 2 electrons.
Hydrogen oxidizing bacteria are often autotrophs (CO_2 fixation) aerobic (O_2 as electron acceptor), forming water and ATP as final products.
Some hydrogen bacteria synthesize two distinct hydrogenases.
Membrane-integrated hydrogenases bind H_2 for use as electron donor in energy metabolism.
Cytoplasmic hydrogenases bind H_2 and directly catalyzes the reduction of NAD^+ to NADH generating reducing power.
Hydrogenases are metal-containing oxidoreductases, whose active site can be either a [NiFe], [FeFe] or [Fe] center.



Ralstonia eutropha

Sulfur compounds oxidation

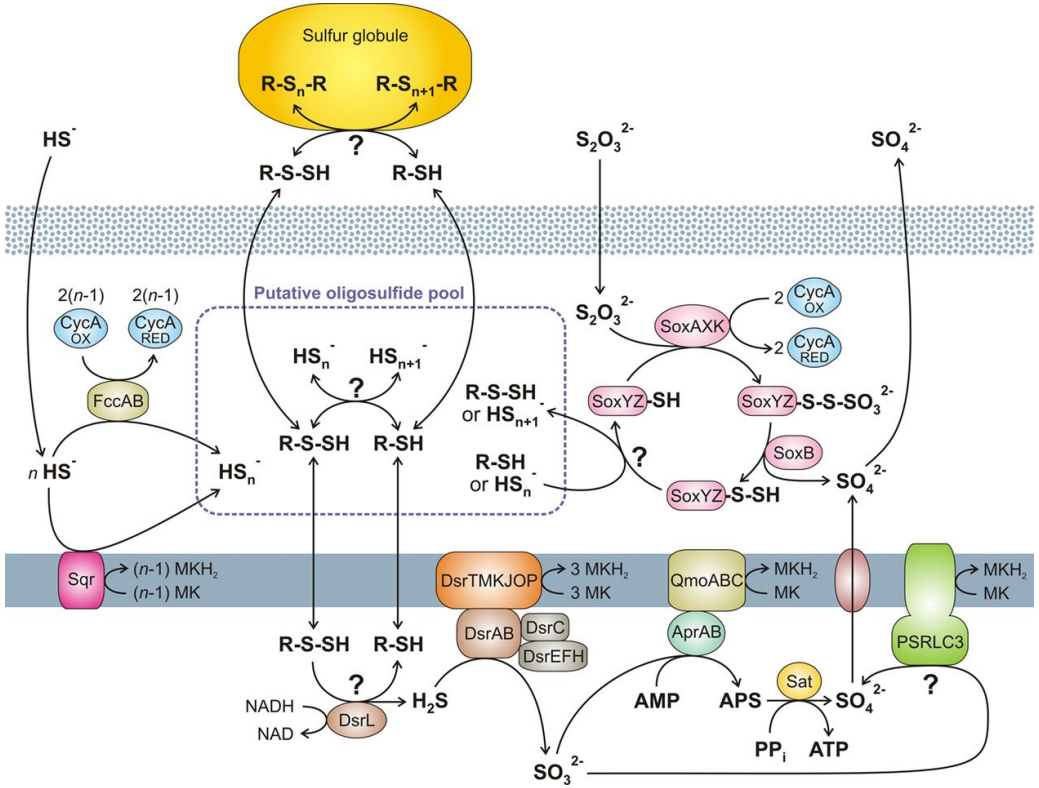
The most common sulfur compounds used as electron donors are hydrogen sulfide (H_2S), elemental sulfur (S^0), and thiosulfate ($S_2O_3^{2-}$).
Sulfur chemolithotrophs are primarily an aerobic group, but some species can grow by anaerobic respiration using nitrate as an electron acceptor. ***Thiobacillus denitrificans*** is a classic example, reducing nitrate to dinitrogen gas.
At least three pathways for sulfur oxidation are known. In two of the systems, the starting substrate is first oxidized to sulfite (SO_3^{2-}). The most widespread system employs the enzyme sulfite oxidase, and the electrons enter the electron transport chain generating a proton motive force that leads to ATP synthesis.
Some sulfur chemolithotrophs, instead, oxidize SO_3^{2-} to SO_4^{2-} via a reversal of the activity of the enzyme adenosine phosphosulfate reductase. In this case, ATP is generated by substrate level phosphorylation.



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Sulfur compounds oxidation

A functionally distinct sulfide and thiosulfate oxidation system has been well described in *Paracoccus pantotrophus*. This system, called the Sox system, consists of more than 15 genes encoding various enzymes and carrier proteins necessary for the **oxidation of thiosulphate directly to sulfate**. It is also present in some green- and purple-sulfur phototrophs, organisms that oxidize sulfide to obtain reducing power for CO_2 fixation rather than for energy storage.



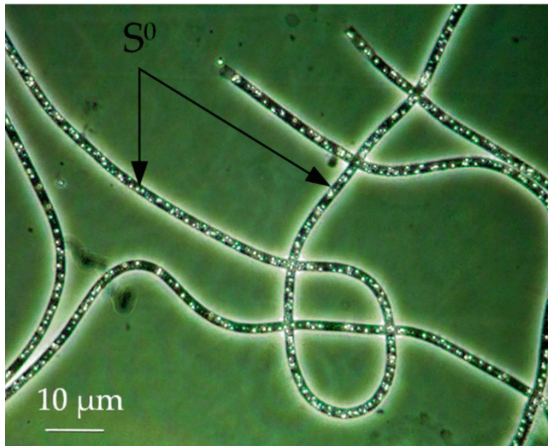
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DOI:10.3389/fmicb.2011.00116

Sulfur compounds oxidation

One of the products of the oxidation of reduced sulfur compounds is H^+. Consequently, as a result, sulfur chemolithotrophs induces acidification of the environment. Because of this, many sulfur bacteria have evolved to be acid-tolerant or even acidophilic. *Acidithiobacillus thiooxidans*, for example, grows best at a pH between 2 and 3.

$$H_2S + 0.5O_2 \rightarrow S^\circ + H_2O$$
$$H_2S + 2O_2 \rightarrow SO_4^{2-} + 2H^+$$
$$S^\circ + 1.5O_2 + H_2O \rightarrow SO_4^{2-} + 2H^+$$
$$SO_4^{2-} + 3H_2 + 2H^+ \rightarrow S^\circ + 4H_2O$$
$$SO_4^{2-} + 4H_2 + 2H^+ \rightarrow H_2S + 4H_2O$$
$$S^\circ + H_2 \rightarrow H_2S$$
$$4S^\circ + 4H_2O \rightarrow SO_4^{2-} + 3H_2S + 2H^+$$

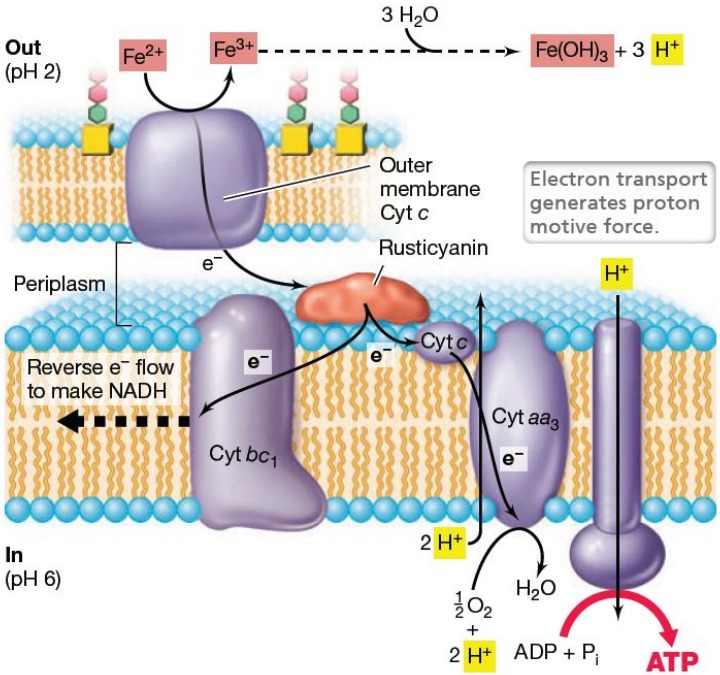


Some sulfide-oxidizing bacteria, such as Beggiatoa, deposit elemental sulfur inside the cell in sulfur granules, where the sulfur exists as a potential energy (electron) reserve. When elemental sulfur is present externally, the organism must attach itself to the sulfur particle because elemental sulfur is rather insoluble.

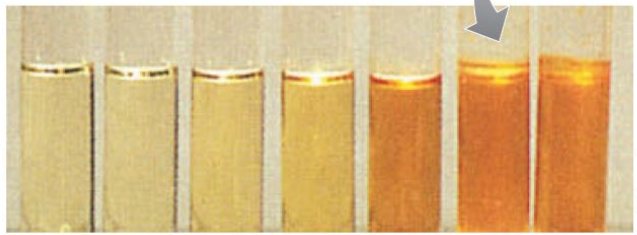
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Iron (II) Oxidation

Iron oxidizers use ferrous iron (Fe^{2+}) as **electron donor**, generally in **aerobic respiration** (O_2 as electron acceptor).
Usually they are chemolithoautotrophic acidophiles, able to grow at pH < 1. This because the ferric iron (Fe^{3+}) produced from the oxidation of ferrous iron (Fe^{2+}) in aquatic environments becomes hydrated to form insoluble ferric hydroxide ($Fe^{3+} + 3H_2O \rightarrow Fe(OH)_3 + 3H^+$) in addition to other iron precipitates, and this drives down the pH. In fact, the best-known iron bacteria, *Acidithiobacillus ferrooxidans*, grows optimally at pH between 2 and 3.
At neutral pH, Fe^{2+} spontaneously oxidized to Fe^{3+}, so opportunities for the iron bacteria in neutral habitats are limited.



$Fe^{3+} + 3\ H_2O \rightarrow Fe(OH)_3 + 3\ H^+$



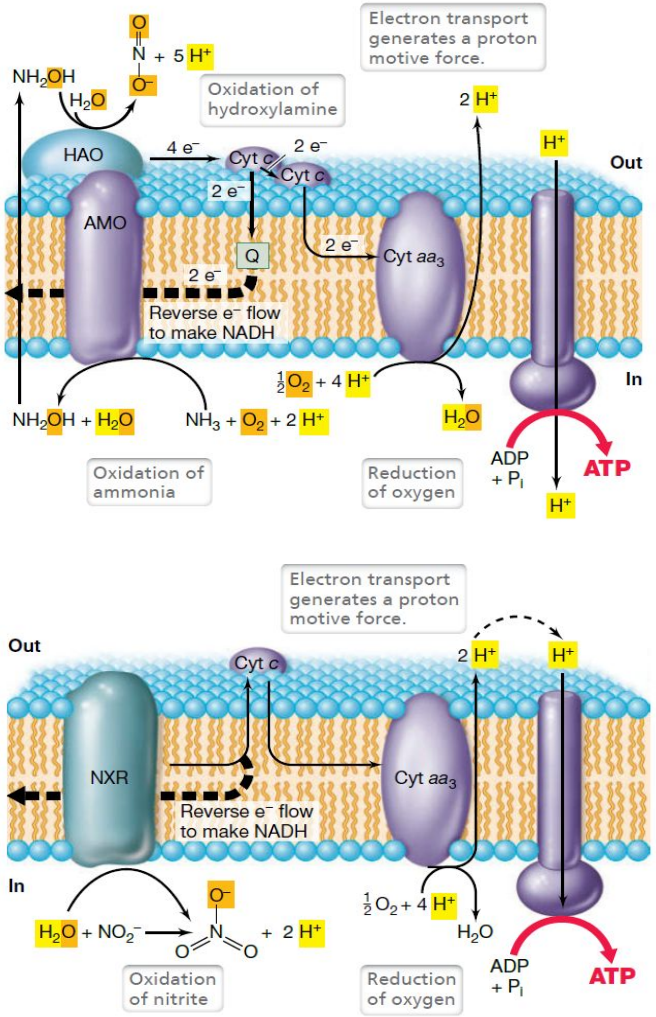
T. D. Brock

University of Naples Federico II

Nitrification

Reduced inorganic nitrogen compounds, such as ammonia (NH_3) and nitrite (NO_2^-) can be aerobically oxidized by the chemolithotrophic nitrifying bacteria. The process of **nitrification** consists of two sets of reactions, generally separately catalyzed by different microbial genera. Bacteria such as Nitrosomonas and Archaea such as Nitrosopumilus oxidize, in two steps, NH_3 to NO_2^- by ammonia monooxygenase and hydroxylamine oxidoreductase, and are therefore known as **ammonia oxidizers**.

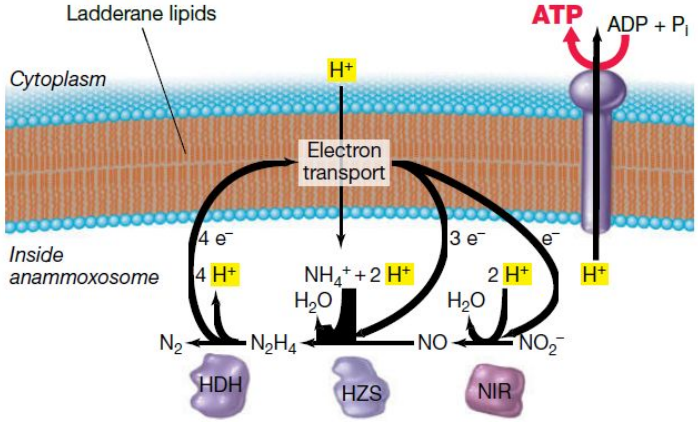
Bacteria such as Nitrobacter oxidize NO_2^- to NO_3^- by the enzyme nitrite oxidoreductase, and we call these organisms nitrite oxidizers or nitrifying bacteria. So far, only bacteria belonging to the genus Nitrospira are reported to catalyze both sets of reactions, oxidizing NH_3 all the way to NO_3^-.

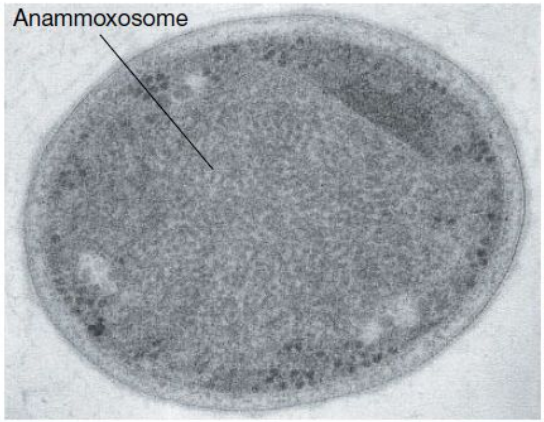


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Anaerobic Ammonia Oxidation (Anammox)

Ammonia can also be oxidized under anoxic conditions by an unusual group of obligately anaerobic bacteria in a process called anammox (anaerobic ammonia oxidation). In this metabolic pathway ammonia is the electron donor for anaerobic respiration that uses nitrite as electron acceptor ($NH_4^+ + NO_2 \rightarrow N_2 + 2H_2O$). Anammox bacteria are also autotrophs, but they fix CO_2 by way of the reductive acetyl-CoA pathway, an autotrophic pathway widespread among some obligately anaerobic autotrophic Bacteria and Archaea.

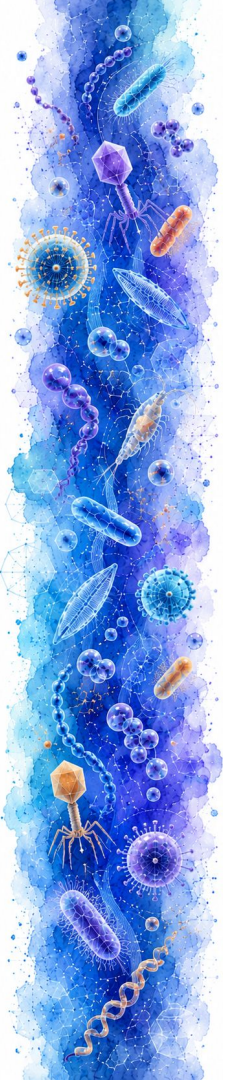




Richard Webb and John A. Fuerst

University of Naples Federico II

Planctomycetes represent an unusual bacterial phylum, with *Brocadia anammoxidans* as representative species, containing in their cytoplasm peculiar membrane- enclosed compartment where the anammox reaction occurs, the **anammoxosome**, whose membranes consist of fatty acids constructed of multiple cyclobutane (C4) rings bonded to glycerol by both ester and ether bonds.



This Week Read

Berg JS, Ahmerkamp S, Pjevac P, Hausmann B, Milucka J, Kuypers MMM. How low can they go? Aerobic respiration by microorganisms under apparent anoxia. FEMS Microbiol Rev. 2022 May 6;46(3):fuac006. doi: 10.1093/femsre/fuac006

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